

18th Mexican Conference on Pattern Recognition

MCPR2026

June 24–27, 2026, CIITA, Ciudad Juárez, Mexico

Program

Wednesday, June 24 (Mexican Central Time UTC-5)	
9:00 – 9:20	Opening Ceremony
9:20 – 10:20	Keynote Address Chair: To be announced <i>New Paradigms in Sensors and Computer Vision Methods + Comments on Artificial Intelligence Production at an International Scale</i> Prof. Walterio Mayol-Cuevas (Department of Computer Science, University of Bristol, United Kingdom)
10:20 – 10:40	COFFEE BREAK
	Session 1: Pattern Recognition and Machine Learning Techniques Chair: To be announced
10:40 – 11:00	Evaluation of Clustering and Consensus Models for Classifying Adulterant Levels in Alcoholic Beverages Using Single-Shot Interferograms E. López-Meléndez, R. I. Álvarez-Tamayo, A. Barcelata-Pinzón, P. Prieto-Cortés, and L. D. Lara-Rodríguez
11:00 – 11:20	Beyond Pareto: A High-Efficiency Approach to Bi-Objective Regression Trees Erick G.G. de Paz, Arturo Hernández-Aguirre, and Iván Cruz-Aceves
11:20 – 11:40	Ensembles of Manifold Learners for Supervised and Unsupervised Learning Fabiola Muñoz Vera, Juan Manuel Pérez Ortega, Eduardo F. Morales, and Hugo Jair Escalante
11:40 – 12:00	Application of Unsupervised Pattern Recognition for the Structural Characterization of Masonry Walls Miguel Ángel Tlatzimatzi Flores, María Guadalupe Medina Barrera, and Eduardo Ismael Hernández
12:00 – 12:20	COFFEE BREAK
	Session 2: Pattern Recognition and Machine Learning Techniques Chair: To be announced
12:20 – 12:40	Clustering Algorithms for Density Variations: An Empirical Comparison Adrián J. Ramírez-Díaz, José Fco. Martínez-Trinidad, and J. Ariel Carrasco-Ochoa
12:40 – 13:00	Dissimilarity-Based Graph Embedding via Local and Global Approximate Graph Patterns Daybelis Jaramillo-Olivares, J. Ariel Carrasco-Ochoa, and José Fco. Martínez-Trinidad
13:00 – 13:20	Comprehensive Validation of Brex Sequential Classifier Model for Robust Pattern Recognition Gerardo Acevedo-Sánchez, J. Pacheco-Senard, A. Alarcón-Paredes, Ó. Camacho-Nieto, and C. Yáñez-Márquez
13:20 – 13:40	A Machine Learning Framework for High-Performance Electrolyte Classification in Lithium Metal Batteries Sergio Rubén Ocampo-Pérez, Noureddine Lakouari, and Outmane Oubram
13:40 – 14:00	Dynamic Assessment of Landslide Susceptibility at Pico de Orizaba Using Ensemble Classifiers and Multitemporal Analysis Adhara Alejandra Avendaño Barajas, L. Altamirano Robles, R. Díaz Hernández, and S. Zapotecas Martínez
14:00 – 15:00	LUNCH
	Session 3: Deep Learning & Neural Networks Chair: To be announced
15:00 – 15:20	Banking Fraud Detection Using Neural Network Models Ostin Uriel Martínez Campos, Andrés Ferreyra-Ramírez, Eduardo Rodríguez-Martínez, and Carlos Avilés-Cruz
15:20 – 15:40	Variational Standard Simplex Hierarchical Encoder for Clustering Ulises Rodríguez-Domínguez
15:40 – 16:00	Chest CT Gender Domain Translation: Comparative Evaluation of CycleGAN and cVAE in Radiometric and Volumetric Coherence I. Delgado Navarro, G. J. Alvarado Flores, L. Altamirano Robles, R. Díaz Hernández, and S. Zapotecas Martínez
16:00 – 16:20	Detection of Dangerous Wildlife in a Dynamic Environment Using Deep Learning David Cruz-Villavicencio and Jesús Alberto Martínez-Castro
16:20 – 16:40	COFFEE BREAK
	Session 4: Deep Learning & Neural Networks/Pattern Recognition and Machine Learning Techniques Chair: To be announced
16:40 – 17:00	Spatiotemporal Prediction of Wildfires Spread Using Deep Learning Ángel Itzcoatl Huizar Bretado, Raquel Díaz Hernández, K. A. Ramírez Gutiérrez, and L. Altamirano Robles
17:00 – 17:20	A Safety-Aware Approach for Automated Urine Culture Screening Using Calibrated Deep Learning G. J. Alvarado Flores, I. Delgado Navarro, R. Díaz Hernández, L. Altamirano Robles, and S. Zapotecas Martínez
17:20 – 17:40	Modeling Tumor Progression in Mammography with Conditional Generative Adversarial Networks Alfonso Rojas-Domínguez and Jesús Yaljá Montiel Pérez
17:40 – 18:00	Customer Preferences Recognition in Neuromarketing: Improving Accuracy with Autoregressive Modeling, Genetic Algorithms, and Naive Bayes Christian Ruiz-Ugalde, R. A. García-Hernández, J. Rojas-Simón, Y. L., and Marco Antonio Ramos-Corchado

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Thursday, June 25 (Mexican Central Time UTC-5)	
9:00 – 10:00	Panel in Spanish <i>Las Voces del Mañana: Niños Genio Debaten sobre Inteligencia Artificial</i> David Da Vinci y Luis Villareal
10:00 – 10:20	COFFEE BREAK
	Session 5: Medical Applications of Pattern Recognition Chair: To be announced
10:20 – 10:40	Machine Learning Algorithms for Identifying Attention-Deficit/Hyperactivity Disorder Processing Electroencephalographic Signals Erika Altair Castro-Verazas, Saúl Brandon Lima-Portillo, Delia Irazú Hernández-Farías, and C. A. Reyes-García
10:40 – 11:00	Graph-Based Modeling of Disfluent Speech for Alzheimer’s Detection Carlos A. Olachea-Hernández, Luis Villaseñor-Pineda, Manuel Montes-y-Gómez, and F. J. Martínez-Santiago
11:00 – 11:20	Factors Associated with Depression in Mexico: An XAI Comparative Analysis of Tree-Based Classifiers Under Imbalanced Data Abigail Romero-Trejo, A. Guzmán-Ponce, R.M. Valdovinos-Rosas, L. Cleofas-Sánchez, and I. Francisco-Valencia
11:20 – 11:40	COFFEE BREAK
	Session 6: Medical Applications of Pattern Recognition / Computer Vision Chair: To be announced
11:40 – 12:00	Hierarchical Incremental Learning for Adaptive Breast Thermography Classification Yareli Aburto-Sánchez, Pilar Gómez-Gil, and Leopoldo Altamirano-Robles
12:00 – 12:20	Automatic Classification of Subcutaneous Mycoses in Clinical Images Using Transfer Learning Vania Déborah Vázquez Palacios, Julio César Pérez Sansalvador, and Humberto Pérez Espinoza
12:20 – 12:40	Upsampling of Sparse Three-Dimensional Point Clouds Using iFactor-KDtree with Adaptive Interpolation Dora-Luz Almanza-Ojeda, Mario-Alberto Ibarra-Manzano, Carlos A. Pérez-Ramírez, and Yair A. Andrade-Ambriz
12:40 – 13:00	COFFEE BREAK
13:00 – 14:00	Keynote Address Chair: To be announced <i>Foundation Models for Microscopy and Medical Data Analysis</i> Prof. Erik Rodner (Department of Data Science and Machine Learning, University of Applied Sciences, Berlin, Germany)
14:00 – 15:00	LUNCH
	Session 7: Language Processing and Recognition Chair: To be announced
15:00 – 15:20	A Two-Stage Textual Preprocessing Pipeline for Medical Image Captioning S. Rascón-Cervantes, G. Ramírez-Alonso, A. Pastor López-Monroy, R. López-Santillán, and N. A. Rendón Mejía
15:20 – 15:40	Semantic Pattern Recognition in Scientific Literature Using Hybrid Topic Modeling and SciBERT With LoRA-Based Validation Yessenia Díaz Álvarez, Raúl Pinto Elías, A. Magadán Salazar, Noé A. Castro Sánchez, and J. Fuentes Pacheco
15:40 – 16:00	Automatic Depressive Symptom Detection on Social Media Using the BDI-II Cielo Aholiva Higuera-Gutiérrez, I. H. López-Nava, M. Montes-y-Gómez, M. Ezra Aragón, and D. E. Losada
16:00 – 16:20	Statistical Analysis of Combined Adversarial Attacks on Spam Detection Models Samantha Acosta Ruiz, Mireya Tovar Vidal, and José A. Reyes-Ortiz
16:20 – 16:40	COFFEE BREAK
	Session 8: Language Processing and Recognition / Computer Vision Chair: To be announced
16:40 – 17:00	InCvT: A Hybrid Inception-Convolutional Vision Transformer Architecture for Speech Emotion Recognition Juan A. Ramírez-Quintana, Eduardo Gallegos-Camarena, A. A. Torres-García, and Verónica Gallegos-Orozco
17:20 – 17:40	Comparative Statistical Analysis of Sentiment Models in Spanish Political Discourse on YouTube Guillermo David Barrera Ortega, María Beatriz Bernabe Loranca, David Pinto Avendaño, and A. Carrillo Canán
17:40 – 18:00	Var Based on Knowledge Transfer from VLMs and Video Descriptions Emilio Vera-Cordero, David Mata Mendoza, Gibran Benitez-García, Hiroki Takahashi, and Mariko Nakano
20:00 – 23:00	CONFERENCE DINNER (Salón ARGOS del Centro de Convenciones Cibeles)

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Friday, June 26 (Mexican Central Time UTC-5)	
9:00 – 10:00	Keynote Address Chair: To be announced <i>Quantum Computing: Foundations, Transition to Industry, and Applications in Pattern Recognition</i> Prof. Salvador Elías Venegas Andraca (Instituto Tecnológico y de Estudios Superiores de Monterrey (ITESM), CDMX, MEXICO)
10:00 – 10:10	COFFEE BREAK
10:10 – 11:25	Panel in Spanish <i>Tecnología con Propósito: Academia, Industria y Gobierno Dialogan sobre Inteligencia Artificial</i> Mtra. Elizabeth Burrola Meléndez (Centro de Innovación e Integración de Tecnologías Avanzadas) Dr. Luis Alfonso Villa Vargas (Centro de Investigación en Computación) More to be announced
11:25 – 11:40	COFFEE BREAK
	Session 9: Computer Vision Chair: To be announced
11:40 – 12:00	Embedded System for Vehicle Environment Perception and License Plate Recognition (LPR) Using Computer Vision and Deep Learning R. Leonardo Méndez-Macías, J. Villegas-Cortez, A. Ferreyra Ramírez, A. Zúñiga-López, and S. Cordero-Sánchez
12:00 – 12:20	Interpretable Human Activity Recognition for Subtle Robbery Detection in Surveillance Videos Bryan Jhoan Cazáres Leyva, Ulises Gachuz Dávila, José Juan González Fonseca, Juan Irving Vásquez, Vanessa A. Camacho-Vázquez, and Sergio Isahí Garrido-Castañeda
12:20 – 12:40	Tree Crown Segmentation in UAV RGB Images via Monocular Depth Estimation Sergio Adán-Juárez, Andrea Magadán-Salazar, J. Fuentes-Pacheco, Raúl Pinto-Elías, and J. Tavira-Villanueva
12:40 – 13:00	UAV Pose Estimation in Low-Light Conditions Using Visual-LiDar Fusion Esteban Tlelo-Coyotecatl, Alejandro Gutiérrez-Giles, and José Martínez-Carranza
13:00 – 13:30	Closing Ceremony
13:30 – 15:00	LUNCH

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Saturday, June 27 (Mexican Central Time UTC-5)			
8:30 – 9:00	Registration		
9:00 – 11:30	Tutorial 1 <i>Understanding Modern Natural Language Processing Techniques: Tokenization, Embeddings, and Transformer Architectures</i> Prof. Humberto de Jesús Ochoa Domínguez Universidad Autónoma de Ciudad Juárez, Ciudad Juárez, MEXICO	Tutorial 2 <i>Neuromorphic Computing for Edge AI: A Practical Introduction with SPICE and Verilog-A</i> Prof. Victor Hugo Ponce Ponce Centro de Investigación en Computación del Instituto Politécnico Nacional, CDMX, MEXICO	Tutorial 3 <i>Robot Modeling and Control</i> Prof. Elsa Rubio Espino Centro de Investigación en Computación del Instituto Politécnico Nacional, CDMX, MEXICO
11:30 – 11:45	COFFEE BREAK		
11:45 – 13:30	Tutorial 1	Tutorial 2	Tutorial 3
13:30 – 14:50	LUNCH		